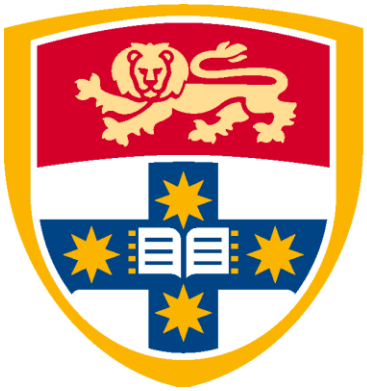


THE UNIVERSITY OF SYDNEY

# ELEC5204 lab report

ELEC5204 Power Systems Analysis and Protection Lab 1: Three-phase Symmetrical Power System and Calculation


| | | |
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# 2. Abstract

The objective of this experiment is to bridge theoretical knowledge with practical application by using MATLAB Simulink to develop a three-phase balanced power system model. Through numerical parameter settings, the circuit is analyzed and the simulation results are verified.

In Task 1, a purely resistive load is implemented to analyze the behavior of a balanced resistive system. In Task 2, the load is modified to a parallel RL configuration, and both active and reactive power are calculated. In Task 3, a more realistic scenario is considered by introducing a non-ideal voltage source with internal impedance connected in series.

The simulation results show good agreement with the manually calculated values of voltage and current. It is also observed that the power factor decreases when the source becomes non-ideal.

This experiment not only enhances the ability to use MATLAB Simulink as a practical engineering tool, but also deepens the understanding of steady-state analysis in balanced three-phase systems.

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# 3. Introduction

Three-phase alternating current (AC) systems are the backbone of modern power transmission and distribution, due to their high efficiency and stability in bulk power delivery. Steady-state analysis of balanced three-phase systems is a fundamental skill for power system engineers, which requires the ability to derive circuit parameters from operating requirements, and validate analysis through simulation. This lab focuses on the modeling and analysis of a symmetrical three-phase power system, The experiment consists of three main tasks: （1）Analyze a purely resistive balanced load and verify its power factor; （2） Design a parallel RL load from given power parameters, and validate the derived component values; （3）Model a non-ideal voltage source with internal impedance, and analyze its impact on circuit performance. MATLAB Simulink with Simscape Electrical toolbox is used to build the circuit model and run simulations. This report presents the detailed calculation process, simulation setup, results and analysis for each task, followed by a conclusion of the lab outcomes.

# 4. system parameters and nomenclature

| Parameter | Definition | Value | Unit |
|---|---|---|---|
| $V_{L-L}$ | Rated line-to-line RMS voltage of the source | $100\sqrt{3}(173.2)$ | V |
| $V_{ph}$ | Rated phase-to-neutral RMS voltage of the source | 100 | V |
| $f$ | System frequency | 50 | Hz |
| $\omega$ | Angular frequency ($\omega=2\pi f$) | 314.159 | Rad/s |
| $R_{line}$ | Per-phase resistance of the transmission line | 50 | Ω |

Table 4-1 Specified System Parameters

| Symbol | Definition | Unit |
|---|---|---|
| $P$ | Active power | W |
| $Q$ | Reactive power | Var |
| $S$ | Apparent power | VA |
| $cos\varphi$ | Power factor | - |
| $R$ | Resistance | Ω |
| $L$ | Inductance | H |
| $X_L$ | Inductive reactance ($X_L=\omega L$) | Ω |
| $Z$ | Impedance | Ω |

Table 4-2 Nomenclature and Symbol Definition

All parameters are kept consistent across all tasks unless otherwise specified:

# 5. task1 :Purely resistive balanced load

## 5.1 task objective

The first part mainly involves using Simulink to model an ideal three-phase voltage source. The load is set as a balanced purely resistive load (with a total active power of 600 W as required, no reactive power), then the equivalent impedance of the load is calculated, the power factor is verified, and finally the simulation results are compared with manual calculations.

## 5.2 Theoretical calculation

### 5.2.1 power factor analysis

For a purely resistive load,the reactive power $Q_{total}=0Var$.

The power factor is defined as the ratio of active power to apparent power:

$$cos\varphi=\frac{P_{toal}}{S_{total}}$$

Where the total apparent power

$$S_{total}=\sqrt{P_{total}^2+Q_{total}^2}$$

substitute$P_{total}=600w$,$Q_{total}=0Var$

$$S_{total}=\sqrt{600^2+0^2}=600VA$$

$$cos\varphi=\frac{600}{600}=1$$

After calculation, the power factor of this system is 1, indicating that the load is a purely resistive load. The current and voltage waveforms have no phase difference and are completely in phase.This eliminates reactive current in the system, improving power transfer efficiency, although resistive losses in the transmission line still exist.

### 5.2.2 Equivalent impedance of the load

(1) split the three-phase power into per-phase power

For a balanced three-phase load, the per-phase active power is 1/3 of the total three-phase active power:

$$P_{ph}=\frac{P_{total}}{3}=\frac{600}{3}=200W$$

(2) derive the load resistance from per-phase power

For a purely resistive load, the per-phase active power follows the formula:

$$P_{ph}=\frac{V_{ph}^2}{R_{load}}$$

Where $V_{ph}=100V$ is the rated phase-to-neutral RMS voltage of the source.

So rearrange the formula: $R_{load}=\frac{V_{ph}^2}{P_{ph}}=\frac{100^2}{200}=50\Omega$

The equivalent impedance of the load is a pure resistance of $50\Omega$ per phase.

### 5.2.3 Peak Values of Load Voltage and Current

Total per-phase circuit impedance:

$$Z_{total}=R_{line}+R_{load}=50+50=100\Omega$$

RMS value of phase current:

$$I_{ph(RMS)}=\frac{V_{ph}}{Z_{total}}=\frac{100}{100}=1A$$

Peak values:

$$V_{ph(peak)}=V_{ph}\times\sqrt{2}=141.42V$$

$$I_{ph(peak)}=I_{ph(RMS)}\times\sqrt{2}=1.4142A$$

## 5.3 simulation setup

① Ideal three-phase voltage source: $100\sqrt{3}V$ line-to-line RMS,50Hz,0phase shift

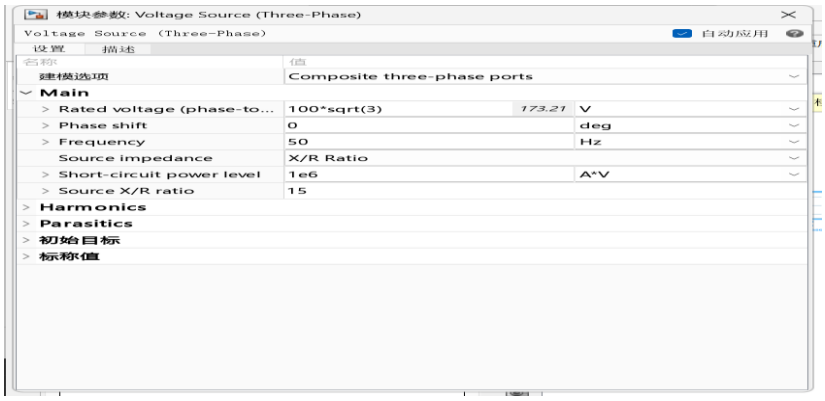



Figure 5-1: Voltage Source Settings.

② Transmission line: purely resistive,R=50Ωper phase

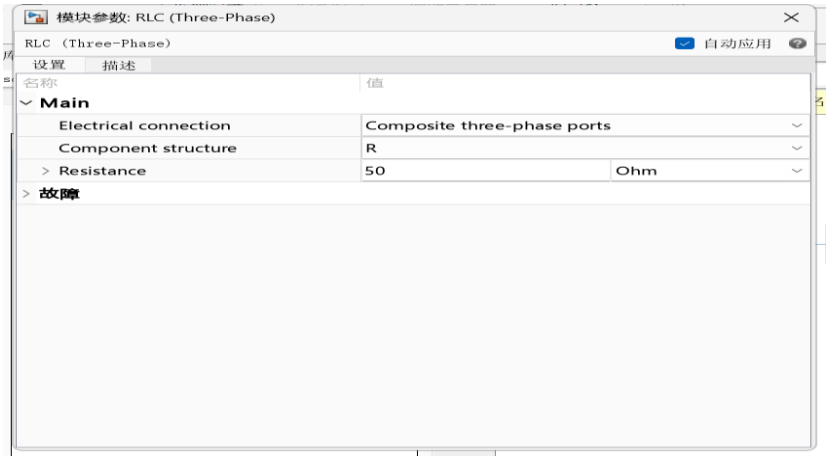



Figure 5-2 :RLC Settings(task1)

③ Y-connected load:

For(task1.1)：Parameterized by rated power,$P$=600$W$,$Q$=0$Var$

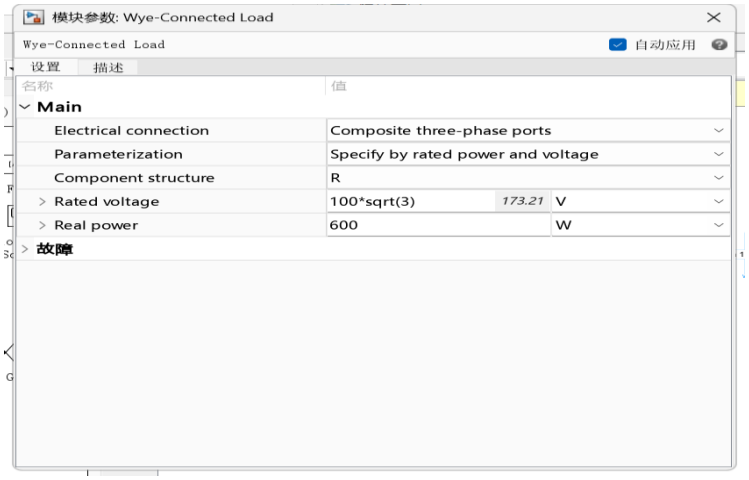



Figure 5-3: Y-connected load Settings（task1.1）.

For(task1.2):parameterized by impedance,$R$=50Ω

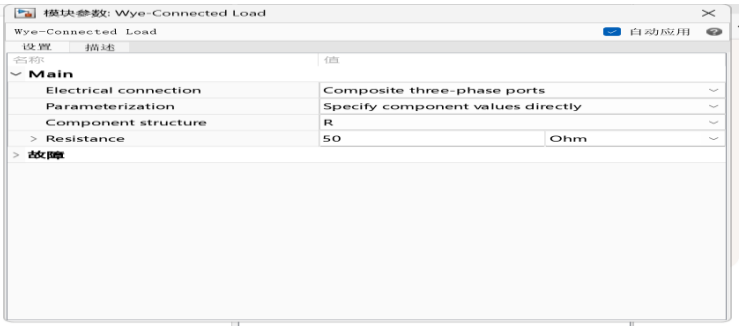



Figure 5-4 :Y-connected load Settings（task1.2）.

④ Three-phase current and voltage sensor: phase-to-ground voltage measurement, SI unit

⑤ Scope: to capture load voltage and current waveforms

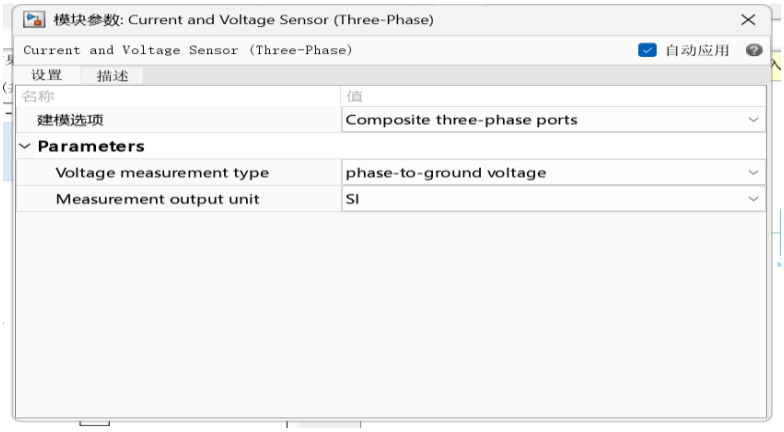



Figure 5-5: Sensor Setup.

## 5.4 simulation result and analysis

### 5.4.1 power factor verification(task1.1)

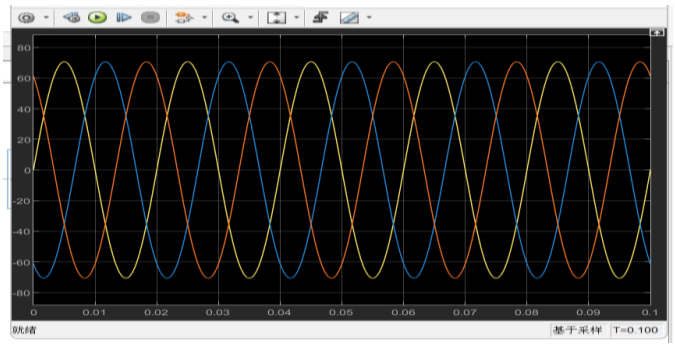



Figure5-6: Load phase voltage waveforms(purely resistive load).

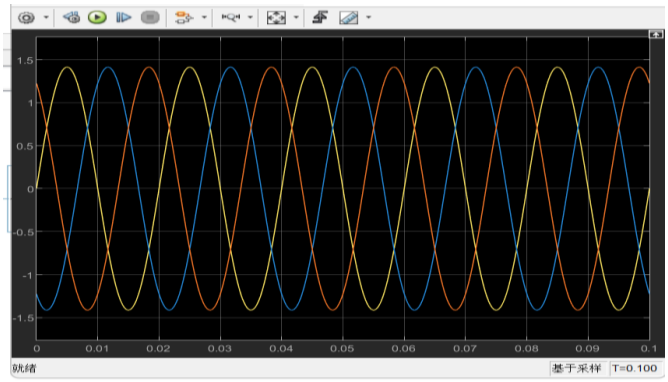



Figure5-7: Load phase current waveforms(purely resistive load).

Analysis:The power factor of the purely resistive load is 1. This is reflected in the simulation results: the voltage and current waveforms are completely in phase, with no phase shift between their peaks and zero-crossing points. No reactive power is consumed in the circuit, so the power factor reaches the maximum value of 1.

### 5.4.2 Equivalent Impedance Validation (task1.2）

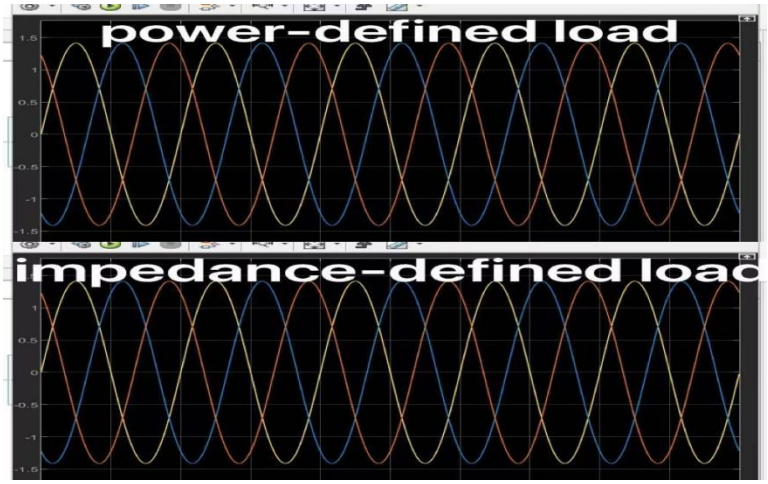



Figure5-8: Comparison of load current waveforms between power-defined load and impedance-defined load in Task 1.

Analysis:The simulation results of the impedance-defined load (R=50Ω) are identical to the results of the power-defined load. This verifies that the derived equivalent impedance is correct, and the two load models are fully equivalent.

### 5.4.3 Cross-Validation of Peak Values (1.3)

Analysis:The peak values of load voltage and current from simulation are 70.7V and 1.414A respectively, which match the manual calculation results closely. This confirms the correctness of the circuit analysis.

## 5.5 Task 1 Conclusion

The purely resistive load has a power factor of 1, the derived equivalent impedance is 50Ω per phase, and all simulation results are consistent with theoretical calculation.

# 6. task2：Parallel RL Inductive Load

## 6.1 Task Objective

This task aims to model an ideal voltage source with a parallel RL load (600W total active power, 381.972Var total inductive reactive power), derive the R and L values of the parallel RL load, verify the power factor, and validate the derived component values through simulation.

## 6.2 Parallel RL Parameters Calculation

### 6.2.1 Split the three-phase power into per-phase power

$$P_{ph}=\frac{P_{total}}{3}=\frac{600}{3}=200W$$

$$Q_{ph}=\frac{Q_{total}}{3}=\frac{381.972}{3}=127.324Var$$

### 6.2.2 Calculate the parallel resistance R

For a parallel RL circuit,the voltage across the resistor and inductor is the same,equal to the phase voltage $V_{ph}$.Only the resistor consumes active power,so:

$$P_{ph}=\frac{V_{ph}^2}{R},$$

Rearrange to solve for R:

$$R=\frac{V_{ph}^2}{P_{ph}}=\frac{100^2}{200}=50\Omega$$

### 6.2.3 Calculate the inductive reactance $X_L$

Only the inductor consumes reactive power in the parallel RL circuit,so:

$$Q_{ph}=\frac{V_{ph}^2}{Q_{ph}}$$

Rearrange to solve for $X_L$:

$X_L=\frac{V_{ph}^2}{Q_{ph}}=\frac{100^2}{127.324}\approx78.54\Omega$

### 6.2.4 Convert inductive reactance to inductance L

The relationship between inductive reactance and inductance is:

$$X_L=\omega L=2\pi fL$$

Where $\omega=2\pi\times50=314.159rad/s$

Rearrange to solve for L:

$$L=\frac{X_L}{\omega}=\frac{78.54}{314.159}\approx0.25H$$

### 6.2.5 Peak Values Calculation(2.3)

Total apparent power :

$$S_{total}=\sqrt{P_{total}^2+Q_{total}^2}=\sqrt{600^2+381.972^2}=711.8VA$$

Power factor:

$$cos\varphi=\frac{P_{total}}{S_{toal}}=\frac{600}{711.8}\approx0.845$$

The load is inductive, so the current lags the voltage by a phase angle of

$$\varphi=arccos（0.845）\approx32.3°$$

Equivalent impedance of parallel load:

$$Z_{load}=\frac{R\times jX_L}{R+JX_L}\approx42.2\angle32.3°\Omega$$

Total per-phase impedance：

$$Z_{total}=R_{line}+Z_{load}\approx88.46\angle14.9\Omega$$

Peak values:

$$I_{ph(peak)}=\frac{V_{ph}}{|Z_{total}|}\times\sqrt{2}\approx1.60A$$

$$V_{load}=I_{ph(PMS)}\times|Z_{load}|\times\sqrt{2}\approx67.4V$$

## 6.3 Simulation Setup

6.3.1 ①Voltage source,transmission line, sensor settings are identical to Task 1

②Y-connected load:

For (2.1):Parameterized by rated power,$P$=600$W$,$Q$=381.972$Var$

Parallel structure.

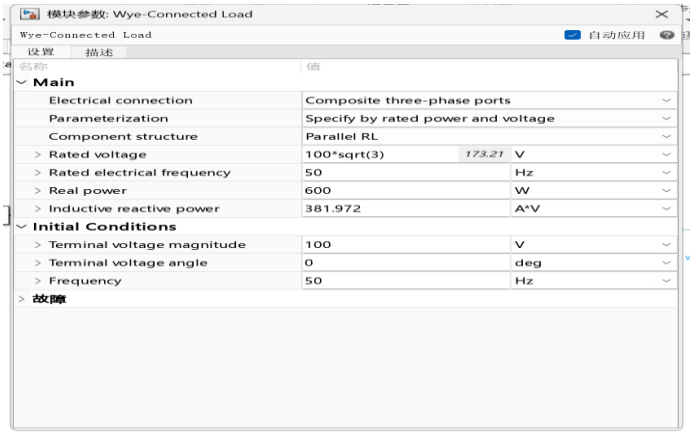



Figure6-1: Y-connected load Settings（task2）

For(2.2):Parameterized by component values,$R$=50Ω，$L$=0.25$H$,parallel RL structure。

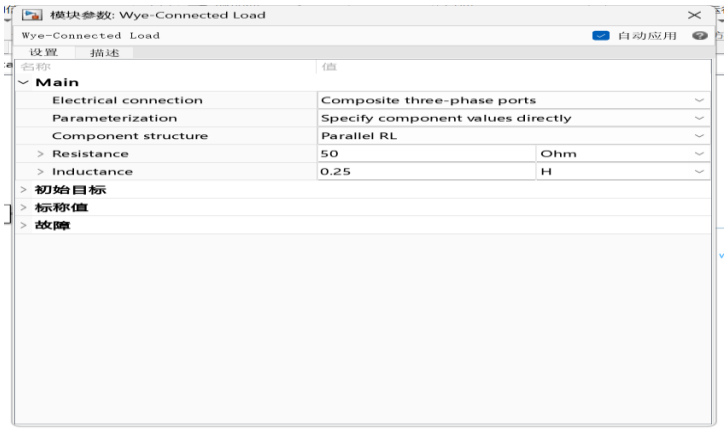



Figure 6-2 :RLC Settings(task2)

## 6.4 Simulation results and analysis

### 6.4.1 Power Factor Analysis(2.1)

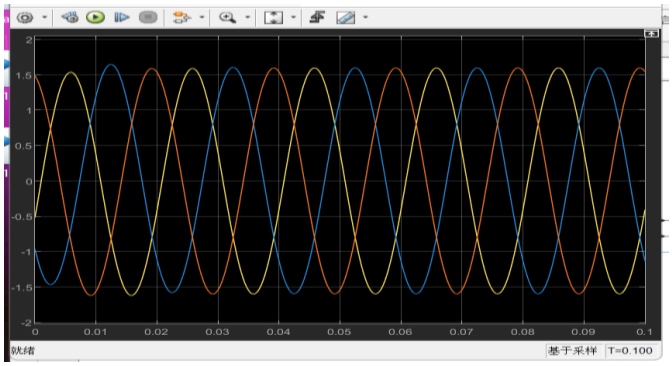



Figure 6-3: Load phase current waveforms for Task 2 (parallel RL load).

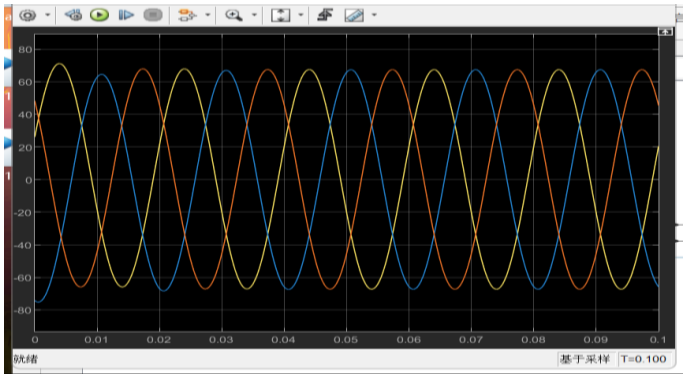



Figure 6-4: Load phase voltage waveforms for Task 2 (parallel RL load)

Analysis:The power factor of the load is 0.845. Compared with Task 1, the current waveform lags the voltage waveform by 32.3°, which is the characteristic of an inductive load. The phase shift between voltage and current reduces the power factor from 1 to 0.845.

### 6.4.2 Parallel RL Parameterized Validation

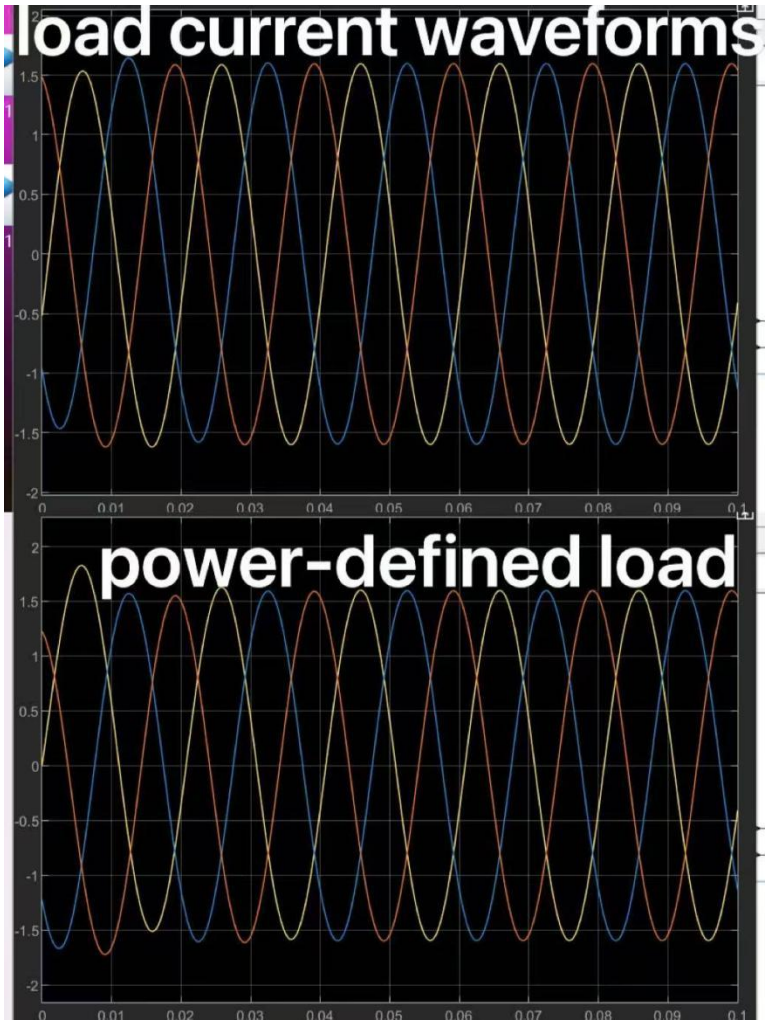



Figure 6-5: Comparison of load current waveforms between power-defined load and parallel RL component-defined load in Task 2.

Analysis:The simulation results of the parallel RL load (R=50Ω, L=0.25H) are identical to the results of the power-defined load. This verifies that the derived R and L values are correct, and the component-defined load is fully equivalent to the power-defined load.

The slight difference in transient disturbance at the initial stage of simulation (0-0.02s) between Task2.1 and Task2.2 is caused by the different initial condition setting logic of the two load parameterization methods, which does not affect the verification of load equivalence. 1. For the load defined by rated power and voltage in Task2.1, Simulink automatically calculates and presets the initial inductor current and magnetic flux according to the rated operating conditions, so the simulation starts

close to the steady state with a smaller transient disturbance. 2. For the load defined by direct component values in Task2.2, the default initial current of the inductor is 0. There is a natural transient process for the inductor current to rise from 0 to the steady-state value, resulting in a more obvious initial disturbance. After 0.02s, the current waveforms of the two modes are consistent within acceptable error in the steady state, with the same amplitude and phase difference. This fully verifies that the parallel RL load with R=50Ω and L=0.25H is equivalent to the power-defined load, which meets the core requirements of the task.

#### 6.4.3 Cross-Validation of Peak Values (2.3)

Analysis:The peak values of load voltage and current from simulation are 67V and 1.6A respectively, which match the manual calculation results closely. This confirms the correctness of the parameter derivation and circuit analysis.

### 6.5 Task 2 Conclusion

The parallel RL load with $R$=50Ω，$L$=0.25$H$meets the given power requirements, with a power factor of 0.845. All simulation results are consistent with the theoretical calculation.

## 7. Task 3:Non-ideal Voltage Source

### 7.1 Task Objective

This task aims to model a non-ideal voltage source, analyze the impact of the non-ideal voltage source on the total circuit power factor, and validate the manual calculation of load voltage and current peak values through simulation.

### 7.2 Theoretical Calculation

#### 7.2.1 Internal Impedance of the Non-ideal Voltage Source (3.1)

Calculate the magnitude of the internal impedance $|Z_s|$

The short-circuit power $S_{SC}$ is the apparent power output by the source when its output terminals are directly short-circuited. When short-circuited, the load impedance

is 0, so the only impedance in the circuit is the internal impedance of the source.The formula for three-phase short-circuit power is:

$$S_{sc} = \frac{3V_{ph}^2}{|Z_s|}$$

Rearrange to solve for $|Z_s|$:

$$|Z_s| = \frac{3V_{ph}^2}{S_{sc}} = \frac{3 \times 100^2}{588.3483} = \frac{30000}{588.3438} \approx 51\Omega$$

Then split the internal impedance into resistance $R_s$and reactance $X_s$

Given the X/R ratio $X_s/R_s = 5$,so $X_s = 5R_s$

The magnitude of impedance follows:

$$|Z_s| = \sqrt{R_s^2 + X_s^2}$$

Substitute $X_s = 5R_s$:

$$|Z_s| = \sqrt{R_s^2 + (5R_s)^2} = R_s\sqrt{1 + 25} = R_s\sqrt{26}$$

Rearrange to solve for $R_s$:

$$R_s = \frac{|Z_s|}{\sqrt{26}} = \frac{51}{5.099} \approx 10\Omega$$

Calculate the impedance reactance:

$$X_s = 5 \times R_s = 5 \times 10 = 50\Omega$$

Last,convert inductive reactance to inductance $L_S$

$$L_S = \frac{X_s}{\omega} = \frac{50}{314.159} \approx 0.1592H$$

### 7.2.2 Peak value of Load Voltage and Current(3.2)

First,calculate the total per-phase impedance

The internal impedance of the source,transmission line resistance,and load impedance are connected in series:

$$Z_{total} = Z_s + Z_{load}$$

Where $Z_s = R_s + jX_s = +j50\Omega$，$R_{line} = 50\Omega$，$Z_{load} = 35.5 + j22.7\Omega$（from task2）

Substitute the values:

$$Z_{load} = (10 + j50) + 50 + (35.5 + j22.7) = 95.5 + j72.7 \approx 120\angle 37.3°$$

Then calculate the peak values

RMS value of the phase current：

$$I_{ph(RMS)} = \frac{V_{ph}}{|Z_{total}|} = \frac{100}{120} \approx 0.833A$$

Peak value of the phase current:

$$I_{ph(peak)} = 0.833 \times \sqrt{2} \approx 1.178A$$

RMS value of the load phase voltage:

$$V_{load(RMS)} = I_{ph(RMS)} \times |\,Z_{load}\,| = 0.833 \times 42.2 \approx 35.2V$$

Peak value of the load phase voltage:

$$V_{load} = 35.2 \times \sqrt{2} \approx 49.8V$$

## 7.3 simulation Setup

Load parameters are identical to Task 2 (parallel RL, R=50Ω, L=0.25H)

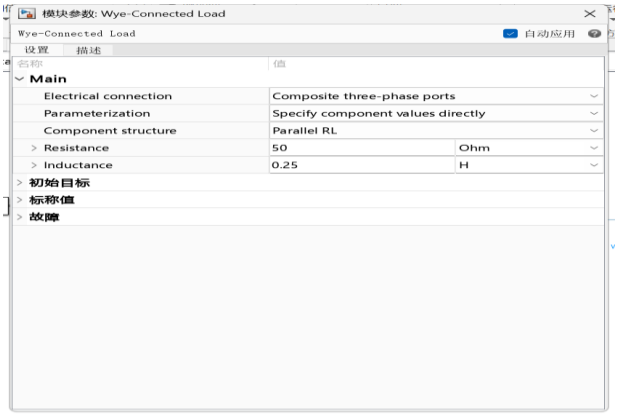



Figure7-1: Y-connected load settings（task3）

a series RLC module is added between the voltage source and transmission line, to model the internal impedance of the non-ideal source: series RL,$R = 10\Omega, L = 0.1592H$,and all other settings are identical to task2.

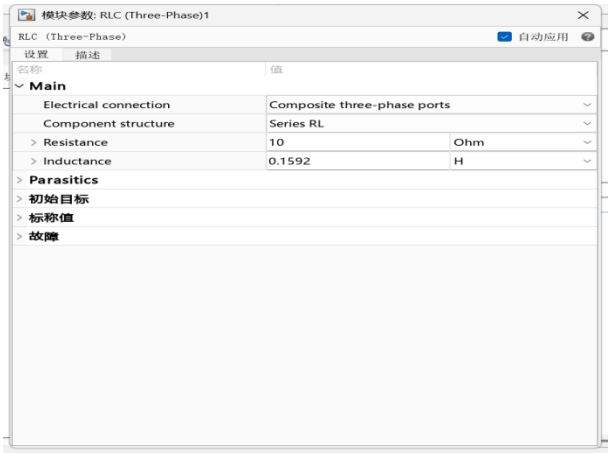



Figure 7-2 :RLC settings for internal impedance of voltage source (task3)

## 7.4 Simulation results and analysis

### 7.4.1 Impact on Power Factor(3.1)

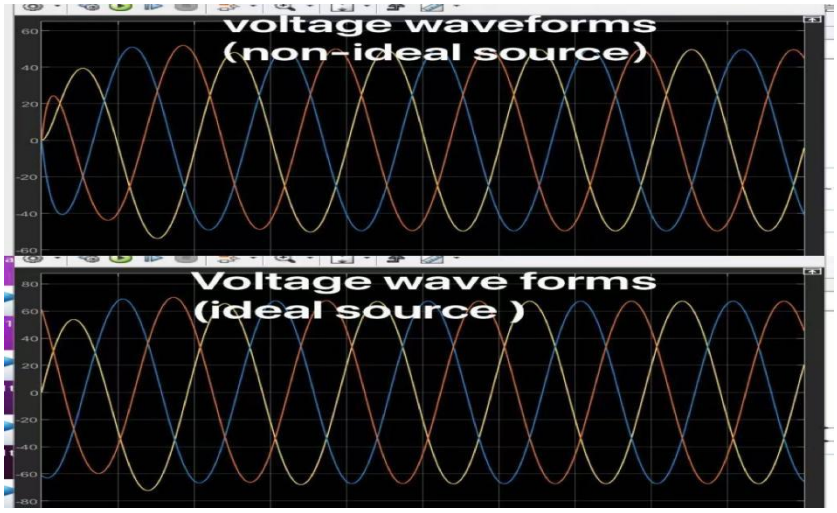



Figure 7-3: Comparison of load voltage waveforms between Task 2 (ideal source) and Task 3 (non-ideal source).

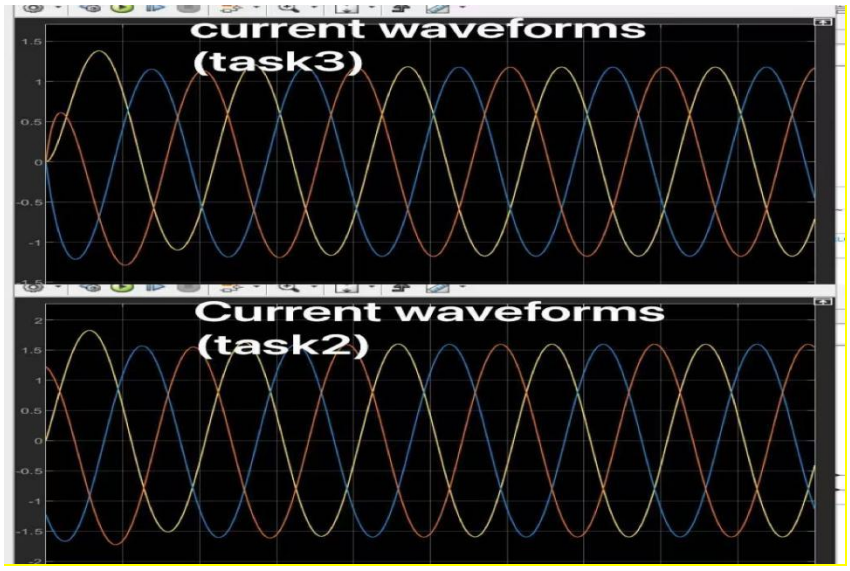



Figure 7-4: Comparison of load current waveforms between Task 2 and Task 3.

Analysis:

Compared with the ideal source case (Task 2), the load voltage peak drops from 67.4V to 49.8V. This is because the internal impedance of the non-ideal source causes an additional voltage drop, which reduces the voltage delivered to the load.

The the overall system power factor decreases: the phase difference between voltage and current increases from 32.3°to 37.3°, because the inductive internal impedance of the source adds extra reactive power to the circuit. This means the non-ideal source will further reduce the power factor of the system, increasing the reactive power loss.

#### 7.4.2 Cross-Validation of Peak Values (3.2)

Analysis:The peak values of load voltage and current from simulation are ~50V and ~1.2A respectively, which match the manual calculation results closely. This confirms the correctness of the internal impedance calculation and circuit analysis.

### 7.5 Task 3 Conclusion

The non-ideal voltage source with internal series RL impedance reduces the load voltage magnitude and the total circuit power factor. All simulation results are consistent with the theoretical calculation.

## 8. Conclusion

This lab completes the modeling, analysis and simulation of a three-phase symmetrical power system under three different load and source conditions. For the purely resistive load, the power factor is verified as 1, with the equivalent impedance derived as 50Ω per phase. For the parallel RL inductive load, the component values R=50Ω and L=0.25H are derived from the given power parameters, and validated by simulation, with a power factor of ~0.845. For the non-ideal voltage source, the internal impedance is calculated, and its impact on load voltage and power factor is analyzed. All simulation results show good consistency with the manual circuit analysis.

# 9. Appendix

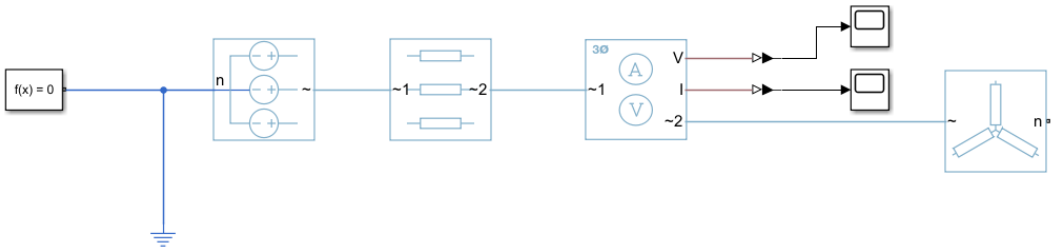



Figure 9-1: Complete Simulink model of the three-phase power system for task1 and task2.

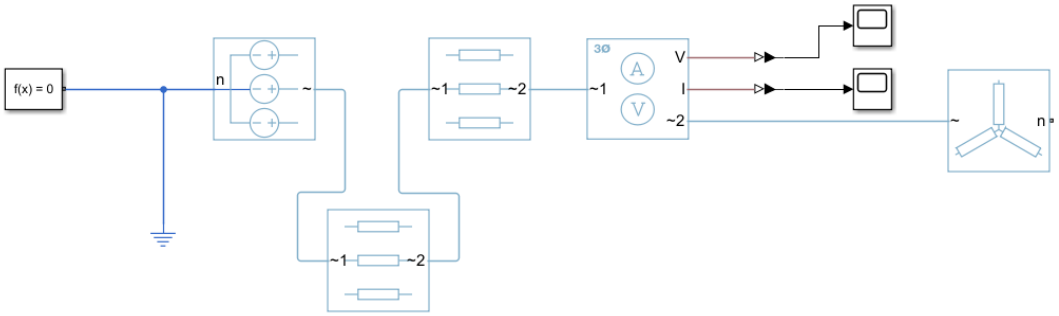



Figure 9-2: Complete Simulink model of the three-phase power system for task3.